JOURNEY

through the Cloud Data World

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Techie Lunch • 2024

Agenda

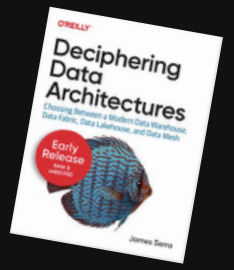
01 Data Architecture Recap

02 "New Age": Data Meshes and Data Producers

03 Solution: Data Lakehouse Architecture in AWS

04 Tidbits & Learning

Data Architecture: A Recap



1980s: Relational DW

Centralized, monolithic OLTP databases. Simple but difficult to scale performance.

2010: Data Lake

Object storage for unstructured data. Scales storage/compute independently; hard to manage.

2011+: Mashups

Introduction of Data Fabric and Modern Data Warehouse architectures.

2020: Lakehouse

Relational abstraction over object stores. Adoption of Delta Lake and Iceberg.

2019: Data Mesh

Decentralized Domain Driven Design approach to organizational data.

Characteristic	Relational data warehouse	Data Lake	Modern Data Warehouse	Data Fabric	Data Lakehouse	Data Mesh
Year introduced	1984	2010	2011	2016	2020	2019
Centralized vs Decentralized	Centralized	Centralized	Centralized	Centralized	Centralized	Decentralized
Storage Type	Relational	Object	Relational and Object	Relational and Object	Object	Domain-specific
Schema Type	Schema-on-Write	Schema-on-Read	Schema-on-Read and Schema-on-Write	Schema-on-Read and Schema-on-Write	Schema-on-Read	Domain-specific
Data Security	High	Low to Medium	Medium to High	High	Medium	Domain-specific
Data Latency	Low	High	Low to High	Low to High	Medium to High	Domain-specific
Time to Value	Medium	Low	Low	Low	Low	High
Total cost of solution	High	Low	Medium	Medium to High	Low to Medium	High
Supported use cases	Low	Low to Medium	Medium	Medium to High	High	High
Difficulty of development	Low	Medium	Medium	Medium	Medium to High	High
Maturity of technology	High	Medium	Medium to High	Medium to High	Medium to High	Low
Company skillset needed	Low	Low to Medium	Medium	Medium to High	Medium to High	High

Data Architectures: Basic Concepts

Data Catalogs

Central store for metadata (types, location, compliance, governance, discoverability).

Batch vs Streaming

- **Batch:** Scheduled ETL jobs (e.g., nightly) for bulk processing.
- **Streaming:** Real-time ingestion and processing of operational changes.

Data Lake Layers

Landing (Bronze)

Raw copy from source; unchanged form.

Transformed (Silver)

Normalized, partitioned, and deduplicated.

Curated (Gold)

Business logic applied; ready for consumption.

"Modern" Ab Initio: Ready for the Cloud?

WHAT IS IT?

- **High Volume:** Software for batch & streaming data applications.
- **Full Ecosystem:** GUI-based development (GDE, MetaData Hub).
- **"Secretive":** Low public profile but used by giants like Sony & Lidl.
- **Old-school:** Traditionally lags behind modern dev & infra practices.

CLOUD MIGRATION

- **Containerization:** Self-contained deployables with CD flow.
- **Kubernetes:** Dispatchers implemented for parallel cluster execution.
- **Challenge:** Achieving true **serverless** behavior for on-demand scaling and cost efficiency.

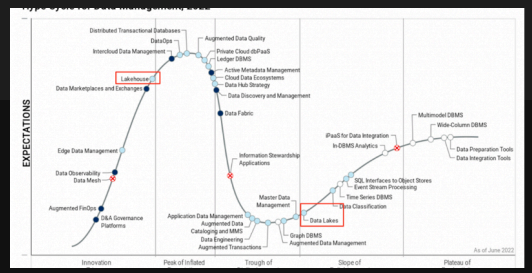
New Age Data: Data Meshes and Data Products

DATA MESH (DDD)

Decentralized approach to data architectures.

- Domain Ownership
- Data as a Product
- Self-Serve Platform
- Fed. Governance

The Reality: Low adoption due to organizational complexity and high technical bar. Often carries antagonistic connotations.



DATA PRODUCT

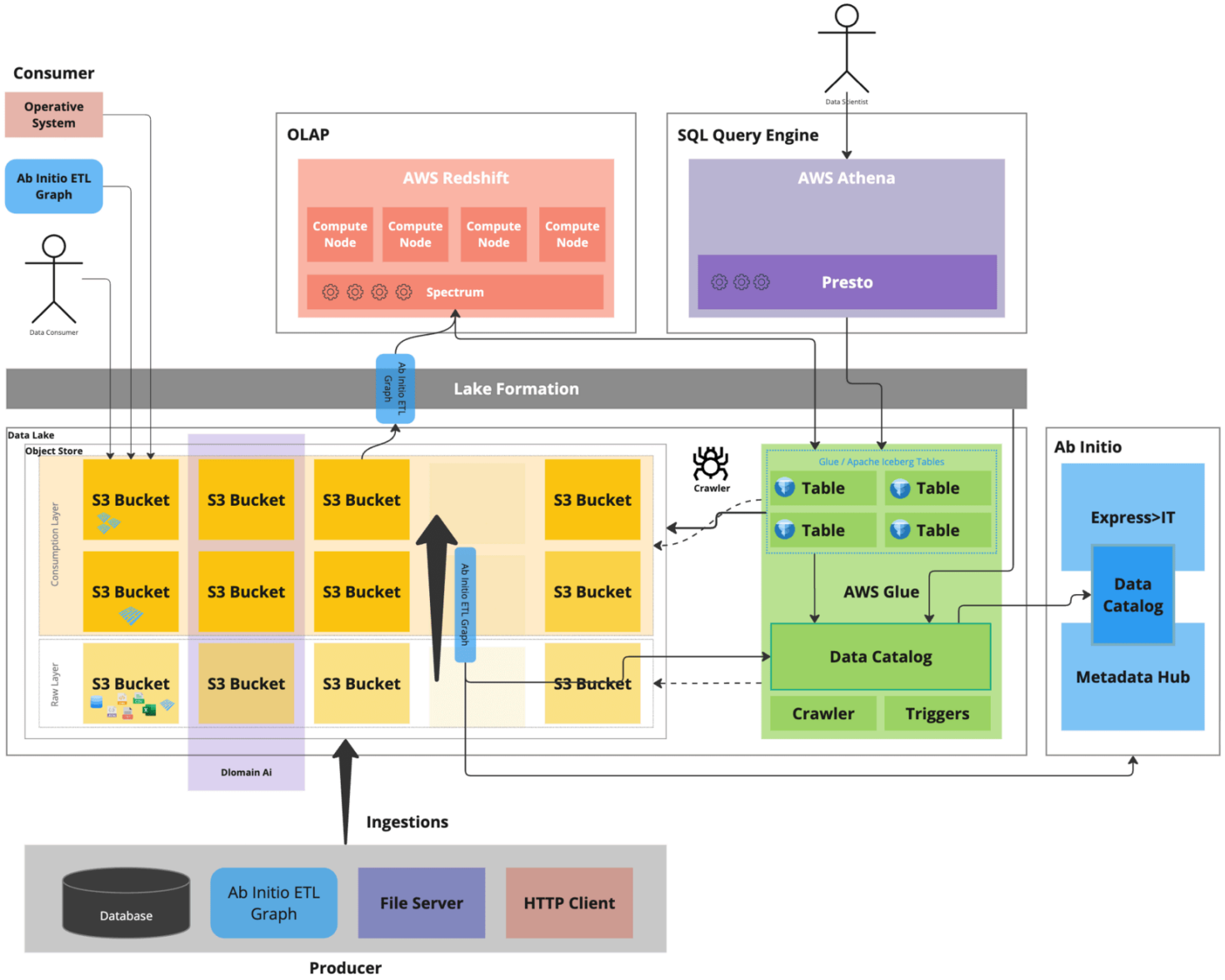
The core concept to be extracted from Data Mesh.

"Focus on the value delivered, not just the pipeline."

RECOMMENDED READING

See latest CTO posts on internal portal.

Data Lakehouse in AWS



Tidbits and Learnings

ORGANIZATION & SKILLS

BI / Analytics departments often have lower average tech skill due to reliance on standard software.

HYPERSCALER ECOSYSTEM

There is a massive, complex ecosystem of data products across major hyperscaler portfolios.

GOVERNANCE & STRUCTURE

- Banking data requires strict central management.
- Data Lakes are notoriously difficult to keep structured.

FUTURE OUTLOOK

Wave of Data Warehouse → "Something Modern" transformations incoming.

THANK YOU!

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Any Questions?